Three-dimensional electronics

https://doi.org/10.1038/s41928-024-01329-3

Building inverters with stacked complementary nanosheet transistors

Xiong Xiong & Yanqing Wu

Developments in the fabrication processes of monolithic complementary field-effect transistors allow inverters with a 48 nm gate pitch to be created.

The demand for integrated circuits with higher device density is driving a shift in device architectures from planar layouts to three-dimensional (3D) stacking. Monolithic complementary field-effect transistors (CFETs) based on silicon nanosheets are a promising architecture for such next-generation logic devices. They are a natural extension of the most advanced current technology – complementary metal–oxide–semiconductor (CMOS) nanosheets – while nearly halving the area required through self-aligned 3D stacking of p-type and n-type nanosheets.

CFET devices have been successfully demonstrated, but the increased complexity and cost of device manufacturing have raised concerns in the semiconductor industry[1]. Achieving industry-compatible CFET technology with power, performance and area advancements will require overcoming several critical process challenges. Reporting at the 2024 IEEE International Electron Devices Meeting in San Francisco, Sandy Liao and colleagues from the Taiwan Semiconductor Manufacturing Company (TSMC) now show that a monolithic CFET process architecture can be used to create a fully operational CFET inverter technology with a 48 nm gate pitch[2].

CMOS logic inverters are fundamental building blocks for large-scale digital logic circuits. They consist of p-type and n-type field-effect transistors (pFETs and nFETs), which are arranged and interconnected horizontally using a series of sequential planar patterning processes. However, in the monolithic CFET architecture, these are vertically stacked, freeing device spacing from the limits of planar patterning, and instead relying on vertical separation. The 3D architecture has area and performance benefits but also introduces additional manufacturing challenges, including interlayer vertical local connections, isolation, independent p-/n-type transistor tuning and layout for signal routing. Building on their previous work[3], the TSMC team addressed three critical process challenges: threshold voltage tuning, vertical metallized drain local interconnects and back-side gate contacts (Fig. 1).

Threshold voltage tuning in vertically stacked CMOS architectures is essential for device performance matching and circuit optimization. The limited physical space within the gate-all-around nanosheet structure makes it difficult to tune the threshold voltage of transistors by the traditional method, where different thickness metal gates with adjustable work function are used, especially when threshold voltages for stacked p- and n-type transistors must be independently controlled. The researchers incorporate vertical dipole patterning technology in the process flow, allowing customized n-dipole and p-dipole patterns to be created for effective work function tuning for the top n-type transistor and bottom p-type transistor, respectively. Thus, they can independently adjust the threshold voltages of pFETs and nFETs on the same wafer without additional physical gate space requirements. This approach achieved a 160 mV reduction in the threshold voltage of the top nFET without affecting the bottom pFET threshold voltage.

In traditional CMOS technology, horizontal metallization can be used for local interconnections, whereas CFETs require middle-of-line local interconnect structures. Liao and colleagues developed a vertical metallized drain local interconnect process that directly connects the top n-type epitaxy and the bottom p-type epitaxy at the common drain, which is validated through electrical measurements. By applying a vertical local interconnection process to the source and drain epitaxy of the top and bottom transistors, the transfer characteristic curve displays the expected bipolar behaviour. By optimizing interconnections in 3D CMOS circuits, the cell area efficiency and overall device performance can be improved.

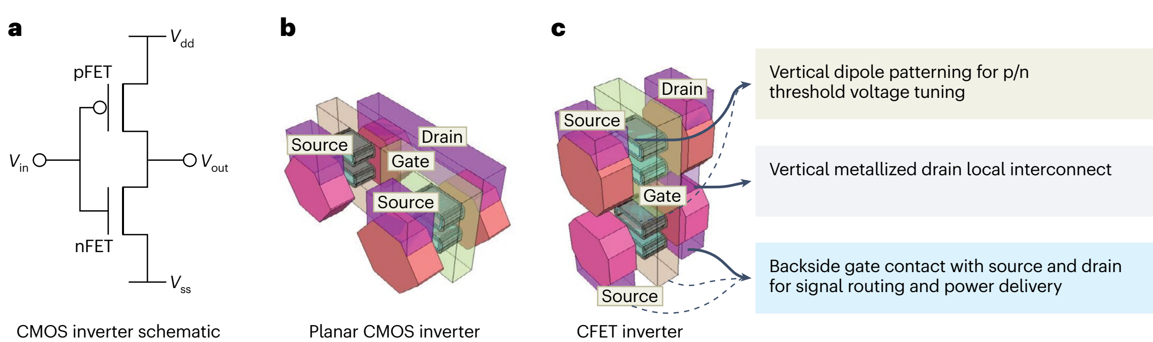

Fig. 1 | Comparison of planar CMOS and CFET inverters based on silicon nanosheets. a, Circuit schematic of an inverter. **b**, Structure of a conventional planar CMOS inverter. **c**, Structure of a 3D stacked CFET inverter, and three key process issues resolved by Liao and colleagues. V_{dd}, power supply voltage; V_{ss}, ground or source voltage; V_{in}, input voltage; V_{out}, output voltage.

CREDIT: SZUYA SANDY LIAO

Addressing problems related to implementing back-side contacts, back-side signal routing and back-side power delivery is essential in order for CFET technology to realize its scaling advantages. After completing front-side processing, the researchers implemented back-side contacts and interconnects in the wafer. This allows the CFET architecture to improve signal and power design flexibility, routing efficiency and the compactness of standard cell layouts when used for large-scale integrated circuits. The pFETs of the bottom layer with back-side contacts achieved a subthreshold swing of about 74 mV dec^{-1} and a drain-induced barrier-lowering value of 47 mV V^{-1}, which closely matches the performance of the nFETs of the top layer with conventional front-side contacts.

These developments in the fabrication process enabled Liao and colleagues to demonstrate a fully functional monolithic CFET inverter on a 300 mm wafer with well-balanced voltage transfer characteristics up to 1.2 V and a 48 nm gate pitch (15 nm gate length), which is a record for monolithic CFETs[4]. The results establish a robust foundation for CFET commercialization. Going forward, further improvements are needed in high-aspect-ratio processes to increase the number of stacked nanosheets, thus providing more flexible effective channel widths. Additionally, optimizing the parasitic effects in vertical structures will be essential to meet the performance, power consumption, area and cost requirements of next-generation logic.

Xiong Xiong & Yanqing Wu
School of Integrated Circuits and Beijing Advanced Innovation Center for Integrated Circuits, Peking University, Beijing, China.
e-mail: yqwu@pku.edu.cn

Published online: 19 December 2024

References

1. Park, J. et al. In *Proc. 2024 IEEE Symp. VLSI Circuits Dig. Tech*. T1.2 (IEEE, 2024); https://doi.org/10.1109/VLSITechnologyandCir46783.2024.10631556
2. Liao, S. et al. First demonstration of monolithic CFET inverter at 48nm gate pitch toward future logic technology scaling. In *Proc. 2024 IEEE Int. Electron Devices Meeting* (2024); https://ieee-iedm.org/program/
3. Liao, S. et al. In *Proc. 2023 IEEE Int. Electron Devices Meeting* 29.6.1–29.6.4 (IEEE, 2023); https://doi.org/10.1109/IEDM45741.2023.10413672
4. Radosavljević, M. et al. In *Proc. 2023 IEEE Int. Electron Devices Meeting* 29.2.1–29.2.4 (IEEE, 2023); https://doi.org/10.1109/IEDM45741.2023.10413678

Competing interests

The authors declare no competing interests.